

Optimal Storage Planning for an Islanded Power System

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Abstract—Currently, the development of microgrids with distributed power sources is receiving growing attention. Integrating both traditional and renewable energy sources into the grid has presented some technical challenges, such as load balancing and frequency regulation. Auxiliary devices like Battery Energy Storage Systems (BESS) are necessary to ensure stable operations and increase the penetration level of renewable energy. However, the cost of investing in BESS systems is relatively high in the present and the future. This article presents a computational model to determine the optimal investment plan for BESS storage in the microgrid system of Bach Long Vi Island, Vietnam. The model is based on the Mixed-Integer Linear Programming (MILP) technique and solved using the PyPSA/CPLEX software. The input data are based on the actual operating data of the studied microgrid system. The multi-period investment framework is tested and its effectiveness is demonstrated by improved economic performance.

Keywords—Microgrid, Energy storage, investment optimization, PyPSA, renewable energy

NOMENCLATURE

$t \in T$	Time intervals (the snapshots).
$g \in G$	Generators include diesel, wind, and solar.
C_g	The operation cost of the generator for technology g (\$/MWh).
C_e^{invest}	The capital cost of extending the storage's inverter power by one MW (\$/MW).
$P_{diesel,min}$	The minimum power output of a diesel generator (MW).
$P_{diesel,max}$	The maximum power output of a diesel generator (MW).
P_{load}^t	The electricity consumption at time t (MW).
P_{BESS}	BESS' inverter rating (MW).
E_{rated}	BESS' capacity rating (MWh).
η	BESS' charging/discharging efficiency.
η_{loss}	BESS' standing losses.
P_g^t	The power output of the generator for technology g at time t (MW).

p_{charge}^t	BESS's charging power at time t (MW).
$p_{discharge}^t$	BESS's discharging power at time t (MW).
u_{diesel}^t	ON/OFF state of diesel generator at time t (binary variable).
u_{BESS}^t	Charging/discharging state of BESS at time t , which equals to 1 if the storage is being charged (binary variable).
e_{BESS}^t	Energy stored in the BESS at time t (MWh).
p_e^{invest}	The expected extended power (MW).

I. INTRODUCTION

Diesel generators have traditionally been the primary energy source in islanded power systems. The development of renewable generation technologies such as wind and solar provides additional primary sources for these systems. Increased utilization of renewable sources is particularly important in the context of the energy transition and reducing carbon footprints in the energy sector. The operational aspect of islanded grid systems is especially challenging because of the relatively small system size and the great fluctuation in load demand. On the other hand, the transition towards more inverter-based generation creates great opportunities for research and development of islanded systems with renewable energy, where they serve as testing grounds for many decarbonization technologies (e.g., grid-forming inverters) before large-scale deployment [1].

In the configuration of microgrid systems, battery energy storage systems (BESS) play an increasingly important role [2]. BESS serves as an energy arbitrage, which helps to maximize the energy output from wind and solar generators and minimize curtailment. Although the cost of BESS installation has gradually reduced, it is still a substantial investment, therefore appropriate optimization models are needed to determine the optimal size and capacity of BESS. The optimization problems are usually formulated as a Mixed-Integer Linear Programming (MILP) [3], [4], heuristic algorithm [5], or a combination of both [6]. In the configuration of microgrid systems, BESS plays an increasingly important role [2]. BESS serves as an energy arbitrage, which helps to maximize the energy output from wind and solar generators and minimize curtailment. Although the cost of BESS installation has gradually reduced,

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Unlike large-scale power system planning, optimization models for islanded systems usually have a lot of variation, depending on specific system configurations. Most studies consider the effect of load growth uncertainty [3], [8]. In [5], the authors consider the effect of low temperatures on the operation of the system components. The effects of shiftable loads and mobile energy storage are considered in [9]. The lifetime characteristics of BESS are considered in [10]. Unlike large-scale power system planning, optimization models for islanded systems usually have a lot of variation, depending on specific system configurations. Most studies consider the effect of load growth uncertainty [3], [8]. In [5], the authors consider the effect of low temperatures on the operation of the system components. The effects of shiftable loads and mobile energy storage are considered in [9]. The lifetime characteristics of BESS are considered in [10].

In this work, we present research to determine the optimal BESS size for an islanded power system in Northern Vietnam. Based on the initial system configuration, this work formulated a multi-investment optimization to determine the best BESS size and investment period so that the system can keep up with the load increase. The research highlights are: i) input data are based on actual operating characteristics (for the wind turbine and the solar farm); ii) a multi-period investment framework is considered. The paper is organized as follows: Section II briefly presents the studied system; Section III presents the optimization framework. The study results are presented in Section IV; Section V concludes the paper.

II. BACH LONG VI ELECTRIC GRID

A. Bach Long Vi microgrid system.

Bach Long Vi island is an offshore district located in the Southeast, over 130km from Hai Phong city; it is the farthest island in the Gulf of Tonkin. In the past, diesel generators were the only source of electrical energy. Bach Long Vi island has great potential for developing renewable energy, especially wind power. In 2021, Northern Power Corporation (EVN NPC) carried out a project that supplied Bach Long Vi island with one new wind turbine, a solar power farm, two new diesel generators, a battery storage system, distribution cabinets, and a microgrid control system. Since then, the island's electricity generation has come primarily from renewable sources, where wind and solar power account for more than 90 percent of the island's electricity demand. The BESS plays a very vital role in the operation of the microgrid because it can provide a grid-forming function, which allows the diesel generator to be turned off in several hours, with decent frequency stability.

Fig. 1 presents the single-line diagram of the current Bach Long Vi (BLV) microgrid. The distributed energy resources (DER) are connected to the microgrid through a low-voltage switchgear and a 0.4 kV/10 kV transformer. BLV microgrid is operated in the islanded mode, which is disconnected from the main grid and supplies the island's local loads.

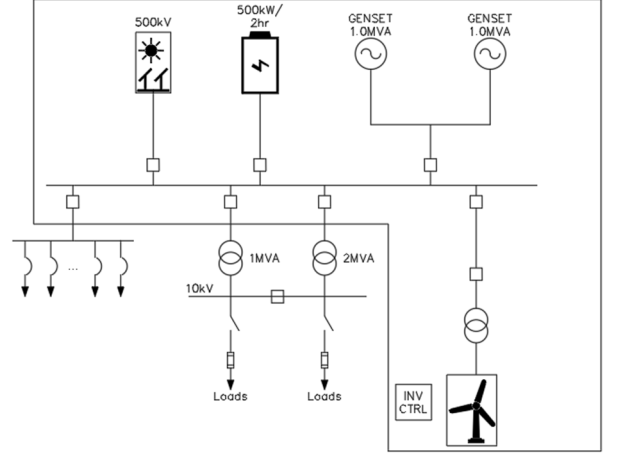


Fig. 1. Simple online diagram of the BLV microgrid.

B. Load profile

Fig. 2 shows the load characteristics for a day. The microgrid's highest load is 400 kW, which is about 80% of the rated capacity of each diesel generator. It is also shown that peak loads occur in the evening.

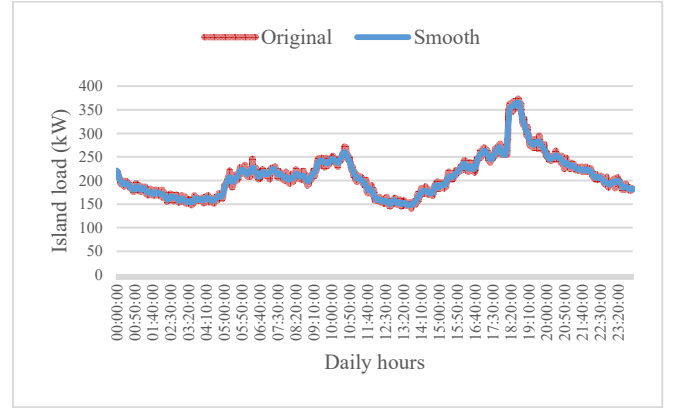


Fig. 2. The load profile in a day.

III. OPTIMIZATION MODEL WITH PYPSA

A. Optimization Model

The island's load demand is expected to increase over the next few decades. Therefore, upgrade plans need to be considered. In particular, the time and amount of BESS capacity need to be correctly determined to keep up with increasing load demand at least cost and using as much as possible wind and solar energies.

This study considers minimizing the total system cost during the operation time, which includes variable and fixed costs of generation and storage with technical constraints. The PyPSA optimization package is used in this work [11].

1) Objective Function

Ignore the fixed costs and start-up or shutdown costs of generators, as well as the fixed costs of energy storage and the inverter of the energy storage. Considering a storage system with zero marginal cost, the objective function is written as follows:

$$\min \sum_t \sum_g C_g \cdot p_g^t \quad [\$] \quad (1)$$

It consists of the dispatch capacities p_g^t of the generator for technology g at time t , with the associated variable cost C_g .

2) Constraints

a) Power Balance Constraint

The total active power output from the diesel generator, the wind plant, the solar power plant and the storage system must equal the total load at any time t .

$$\sum_g p_g^t + p_{discharge}^t - p_{charge}^t = P_{load}^t \quad [MW] \quad (2)$$

b) Constraints of the generators

The dispatch of a diesel generator unit cannot be less than the minimum capacity and greater than the maximum capacity. This technical constraint is described as follows:

$$u_{diesel}^t \cdot P_{diesel,min} \leq p_{diesel}^t \leq u_{diesel}^t \cdot P_{diesel,max} \quad [MW] \quad (3)$$

In (3), the binary variable u_{diesel}^t which indicates whether the generator is running (1) or not (0) at time t .

The power output of the wind and solar power systems has to be smaller than the available power output of each unit at time t .

$$0 \leq p_{wind}^t \leq P_{wind}^{max,t} \quad [MW] \quad (4)$$

$$0 \leq p_{solar}^t \leq P_{solar}^{max,t} \quad [MW] \quad (5)$$

c) Storage Constraints – inverter capacity

The following constraints show that the BESS charging/discharging power is limited by the inverter's power rating:

$$0 \leq p_{charge}^t \leq u_{BESS}^t P_{BESS} \quad [MW] \quad (6)$$

$$0 \leq p_{discharge}^t \leq (1 - u_{BESS}^t) P_{BESS} \quad [MW] \quad (7)$$

The binary variable u_{BESS}^t ensures that at each point in time, the BESS can be operated in only one status: charging or discharging.

d) Energy capacity constraint

Each storage system has only a fixed energy storage capacity. The storage should not be overcharged, so the energy level at time t must be less than the rated energy capacity:

$$0 \leq e_{BESS}^t \leq E_{rated} \quad [MWh] \quad (8)$$

e) Constraint related to the process of charging/discharging of the storage device

The BESS state of charge can be determined by the following model, where η is BESS charging/discharging efficiency:

$$e_{BESS}^t = (1 - \eta_{store}) e_{BESS}^{t-1} + \eta p_{charge}^t - p_{discharge}^t / \eta \quad (9)$$

[MWh]

The storage unit has a standing loss η_{store} (self-discharging leakage rate). In this study, the initial state of charge is set to the final state of charge for the group of periods t .

B. Multi-investment period

This paper investigates the long-term capacity expansion planning of the storage system in the islanded power system. The project's total costs include both the expansion cost of the storage system, the inverter and the operational cost of the grid, using a multiple investment model. The model optimizes the Net Present Value (NPV) of the investment cost over several years in which the asset is active (throughout its lifetime after the build year). For simplicity, this study assumes that the charge/discharge duration of the BESS is fixed; that is, regardless of the time of investment, the installed BESS always has a rated capacity proportional to the rated charge/discharge power at a given factor. Then, the BESS investment cost function can be simply expressed as a function of P_e^{invest} .

The annualized investment costs in each period are always considered and the particular investment cost C^{invest} [USD] is expressed by:

$$C^{invest} = \sum_{a=0}^{A-1} \frac{1}{(1+r)^a} \left(\sum_{e|b \leq a \leq b+L} C_e^{invest} \cdot P_e^{invest} \right) \quad [\$] \quad (10)$$

where A is the investment period and e is the period index number of expansions for the energy storage system in A . P_e^{invest} represents the new generation power built for the BESS-inverter with build year b , lifetime L . If the BESS is expanded in year b , it's available from 1st January in year b up to 31st December in year $b + L - 1$. The discount factor $(1+r)^{-a}$ represents the discounting cash flows r year after the present for interest rate r .

Taking into account the operation costs of the generators in the microgrid, the objective function of the optimization problem can be formulated as follows:

$$\min \sum_{a=0}^{A-1} \frac{1}{(1+r)^a} \left(\sum_{e|b \leq a \leq b+L} C_e^{invest} \cdot P_e^{invest} + \sum_t \sum_g C_g \cdot p_g^t \right) \quad [\$] \quad (11)$$

IV. RESULTS

A. Input data

1) Microgrid

This study is based on the actual power system at Bach Long Vi Island, Hai Phong City, Vietnam. The power grid was started and installed in 2020 with equipment and specifications listed in TABLE I.

TABLE I. INPUT PARAMETERS

PARAMETER	VALUE
Diesel generator rated power	0.5MW
PV system installed capacity	0.5MW
Minimum PV generation capacity	0MW
Wind turbine installed capacity	0.5MW
Minimum wind turbine generation capacity	0MW
BESS capacity	2MWh

Maximum charge/discharge power of BESS	0.5MW
BESS charge/discharge efficiency	96%
BESS standing losses coefficient	1%
Diesel generator variable cost	0.25\$/MWh

2) Solar and wind parameters

a) Wind turbine characteristics

The power-wind speed characteristic represents the relationship between wind speed and the available power output of the wind turbine. This curve is constructed for planning, prediction, monitoring, and optimization purposes. Wind turbines with the same nameplate capacity may produce different power outputs at the exact location and wind speed. This work estimates the wind turbine power curve based on its historical operational data. Different models have been proposed in the literature to build the most accurate mathematical relationship between wind turbine power and wind speed. Besides, the accuracy of each model is evaluated via the R-squared (R^2) value. It shows that the model using a third-degree polynomial equation achieves the highest accuracy, with an R-squared value of 98.6% [12].

Based on the data of wind speed and wind turbine power at Bach Long Vi Island from August 1, 2022, to March 11, 2023, using the nonlinear least squares method written by the Python programming language, the obtained result is the characteristic curve equation: $f(v) = -4.08v^3 + 82.88v^2 - 469.08v + 864.83$. Thus, the characteristic curve of the wind turbine is:

$$P_{wind}^{max,t} = \begin{cases} 0 & v \leq 4 \\ f(v) & 4 < v \leq 10 \\ 500 & v > 10 \end{cases} \quad (12)$$

The calculated squared error is $R^2 = 0.9564$ and the characteristic curve is shown in Fig. 3.

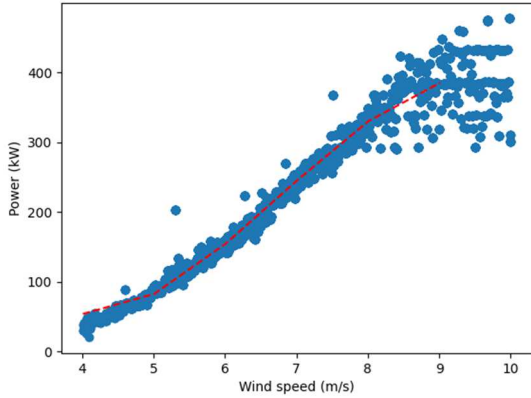


Fig. 3. Wind turbine characteristics, based on wind speed.

b) Photovoltaic characteristics

In photovoltaic cells, the solar cell module's power depends on the sun's radiation, which affecting the operation of the photovoltaic cell. When the ambient temperature remains constant, with an increase in solar radiation, the open-circuit voltage increases insignificantly, and the short-circuit current increases rapidly in response to the increased radiation. Consequently, the output power of the PV module increases. Solar radiation is a factor that directly affects the electrical power output of PV modules.

Therefore, based on the Global Horizontal Irradiance (GHI) data obtained at Bach Long Vi Island, the authors

construct the PV characteristic curve between the solar radiation intensity and the module's output power. By using the nonlinear least squares method supported by the Python programming language, the obtained result for the PV characteristic curve is as follows (Fig. 4):

$$P_{solar}^{max,t} = \begin{cases} 0 & i = 0 \\ 0.46i + 15.09 & i \leq 750 \\ 0.10i + 276.06 & i > 750 \end{cases} \quad (13)$$

The R-squared value is $R^2 = 0.9413$.

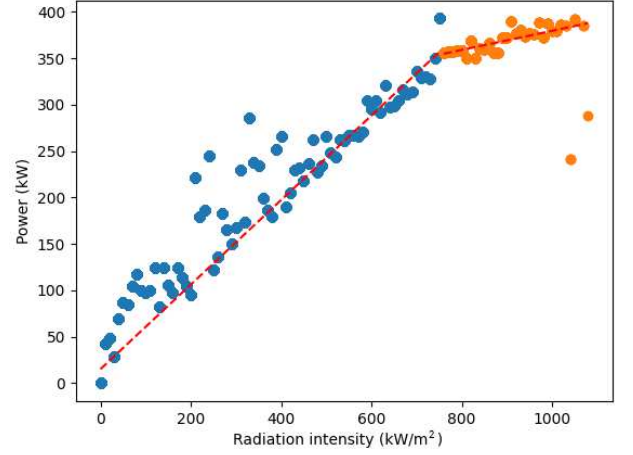


Fig. 4. PV characteristics according to radiation intensity.

3) Investment cost of the storage system

Lithium battery technology is considered in this work because it is currently used in the island's BESS. According to a recent MIT study [13], Li-ion storage will likely remain the preferred technology for short-duration applications. According to the latest report from Bloomberg New Energy Finance (BNEF) [14] and other relevant sources, the prices are expected to decrease gradually (albeit a slight increase in 2022); there is a high possibility of a 10% reduction within the following 12 months. Based on that, we can roughly predict the battery price in 2041 (Fig. 5):

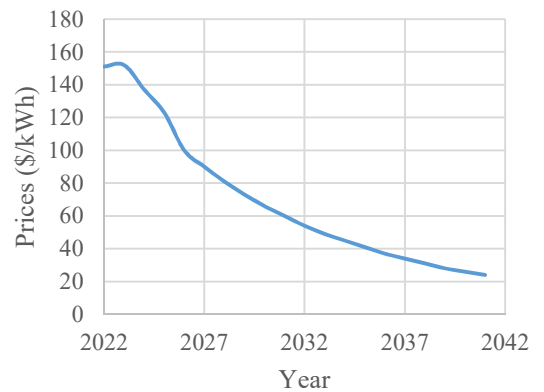


Fig. 5. Li-ion battery investment rate over the years.

B. Optimization results

The optimization model is formulated in combination with the CPLEX solver supported by PyPSA. This MILP problem is solved on a 2.3 GHz Intel(R) Core(TM) i7-1051U CPU with

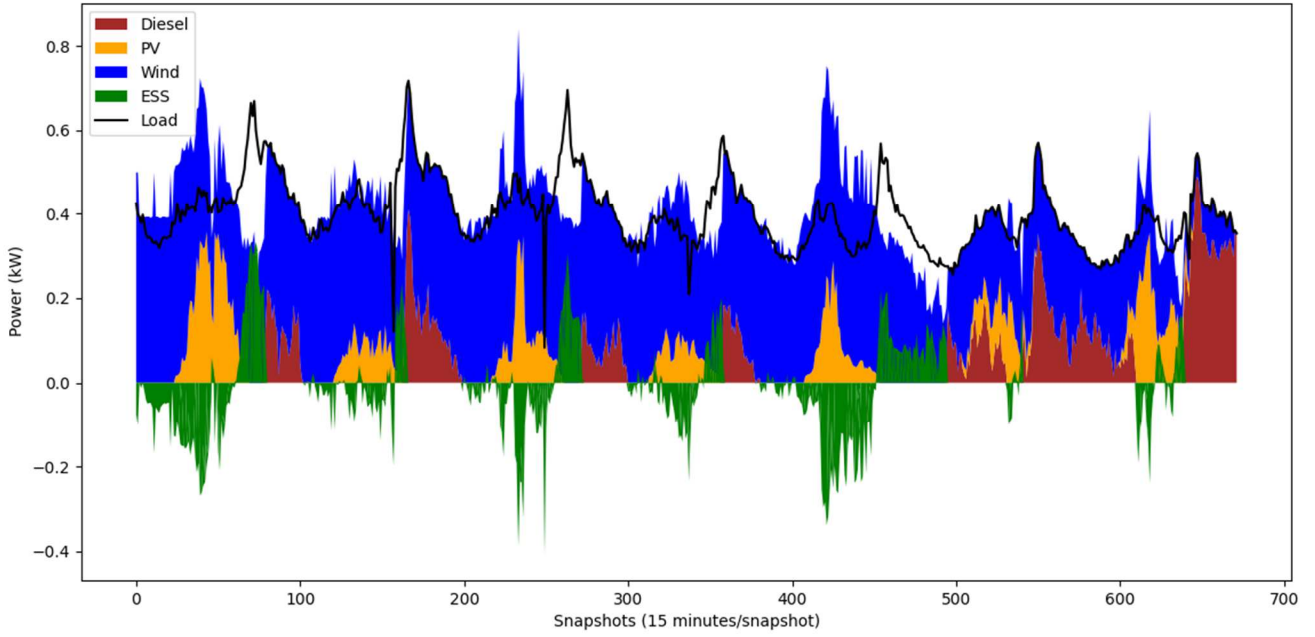


Fig. 6. Bach Long Vi microgrid's optimal weekly schedule with BESS investment in 2040.

8GB RAM, that was using CPLEX version 22.1.1 in a Python programming environment. The simulation uses historical data from the island and forecasts load growth through an annual growth rate, with a total of 23,040 snapshots over a 20-year period. After the optimal investment period of the storage system, the microgrid schedule for a typical week in 2040 is represented in Fig. 6.

TABLE II. shows the operating cost over the 20-year horizon when no new installation is carried out (i.e., the diesel generators will be the balancing source when needed).

TABLE II. OPERATING COSTS WITH VARYING ANNUAL LOAD GROWTH RATES.

Load Growth Rate	0.03	0.04	0.05	0.06	0.07
Operating Cost (M\$)	3.1	3.71	4.38	5.11	5.92
Electricity produced from diesel (MWh)	7745	9272	10950	12774	14794

As seen in TABLE II. , the operating costs of the power grid are quite substantial: there is significant diesel consumption. Evaluating the investment efficiency and combining the BESS investment rate, we consider several investment scenarios over a 20-year period with different timeframes:

- Scenario 1: Investment in 2022 with an unconstrained capacity expansion.
- Scenario 2: Investments every 5 years (investments in 2027, 2032, and 2037) with unconstrained capacity expansion.

TABLE III. SIMULATION RESULTS FOR EACH SCENARIO WITH A 3% LOAD GROWTH RATE.

	Objective function (M\$)	Electricity produced from diesel (MWh)
Scenario 1	2.07	6798
Scenario 2	1.67	5926

From TABLE III. , it can be observed that Scenario 2 investment yields higher efficiency compared to Scenario 1. The diesel generation capacity is lower, and the integration of renewable energy is more successful. When comparing with the non-investment case from TABLE II. , we can see a significant reduction in operating costs. Because of the declining investment cost of Li-ion batteries, incremental upgrades prove to be more effective than the one-time investment approach.

TABLE IV. SIMULATION RESULTS FOR SCENARIO 2 WITH EACH LOAD GROWTH RATE.

Load growth rate	P_e^{invest} 2027 MW	P_e^{invest} 2032 MW	P_e^{invest} 2037 MW	Objective Function M\$
0.03	0.984	2.262	2.725	1.67
0.04	1.098	2.350	2.557	1.97
0.05	1.214	2.277	2.277	2.33
0.06	1.229	2.150	2.30	2.75
0.07	1.229	1.900	2.701	3.24

TABLE IV. shows the expanded capacity over the investment years by increasing the rated power $P_b^{nom}(b)$: build-year) with each load growth rate, and the objective function includes both investment and operation costs of the power grid. Comparing it $P_b^{nom}(b)$ with varying load growth rates. Comparing these results with the cost of operation without investment, the effectiveness is evident, where the cost can be reduced by approximately 50%. Regarding the investment capacity of the additional expansion over 20 years, it is about 8 MWh (increasing the rated power

limit by an additional 2MWh and the maximum charging time of the battery is 4 hours).

C. Sensitivity Analysis

TABLE V. shows optimization results with different values of battery efficiency. It can be seen that as the BESS efficiency decreases, the installed battery size tends to be generally smaller while the optimal operating cost increases. In detail, with the same investment period of 2032, in the case of the battery efficiency being 96%, the BESS' optimal installed capacity is 2.262 MW, while this value is only 1.907 MW when the BESS efficiency drops to 88%. This is because the investment cost of the battery is quite high, requiring a high return on capital each year to attract investment. The reduced battery efficiency will decrease the battery's profit from energy arbitrage and then reduce investors' profit expectations. Consequently, the investment in installing batteries will be reduced.

TABLE V. SIMULATION RESULTS WITH BATTERY PERFORMANCE VARIATION

Battery efficiency	P_{invest}^{2027} (MW)	P_{invest}^{2032} (MW)	P_{invest}^{2037} (MW)	Objective Function (M\$)
0.96	0.984	2.262	2.725	1.67
0.94	1.004	2.250	2.689	1.68
0.92	0.963	2.185	2.690	1.70
0.90	0.908	2.056	2.553	1.72
0.88	0.925	1.907	2.572	1.73

V. CONCLUSIONS

The paper presented a multi-period investment optimization for an actual microgrid system with renewable generation sources and battery energy storage. The model takes into account the cost of investing in BESS systems, the actual operating data of the studied microgrid system and the multi-period investment framework. The multi-period investment framework allows the network assets to be upgraded in multiple stages.

The results show that investing in BESS reduces the consumed diesel generation capacity and increases the penetration level of renewable energy in the microgrid. Compared to the system without BESS, the total system cost is also optimized, leading to better economic performance. Moreover, this is attributed to the decreasing cost of Li-ion batteries, which makes gradual upgrades more effective than a one-time investment approach. This study's framework has the potential to be utilized in other microgrid systems that possess similar characteristics.

In future works, the authors will consider other types of storage technologies, such as those based on hydrogen, vanadium redox flows, etc.

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